wealth (or more correctly how wealth is perceived by the investor) that matters; it is wealth relative to a reference point that is important. With habit utility, bad times are defined by an investor's wealth coming close to habit.[23]

Agents quickly get used to a particular level of consumption. When you're stuck at the back of the plane, you're just glad that you're travelling somewhere exotic; the hard seat, having your knees crammed into the tray table in front of you, and battling your neighbor over the armrest aren't that bothersome (in most cases). Then you get an upgrade to business class. Whoa, is it hard to go back! And then you get the lucky bonanza of a first-class upgrade on one of those spiffy international airlines whose pampered patrons enjoy seats that lie flat, plus unlimited champagne. Stepping back down to cattle class really hurts, even though by any objective standard you are better off—and certainly no worse off—than had you never escaped from coach.

The "habit" in habit utility can be interpreted as a "subsistence" level of required wealth—perhaps literally subsistence consumption, but more generally it is the level required to live a certain lifestyle. As wealth comes close to habit, the investor acts in a much more risk-averse fashion. Thus, with habit utility, *risk aversion is endogenous*. There is an overall curvature of an agent's utility function, but local curvature is always changing.

Habit, too, evolves over time. Your income could skyrocket, so you really don't have to go back to economy class. Or your investment bank employer could blow up, and then you are stuck at the back of the plane. Habit can be external—driven by factors outside an investor's control, such as macro factors—or internal, where it depends on the past consumption and wealth of an individual.

Asset allocation with habit utility requires you to state your portfolio's returns in relation to your habit. In the case where you are close to habit, the asset owner's implied risk aversion will be very high—this is a bad time for her and she wants safe assets. But in good times when her wealth is far above habit, she will be risk tolerant and hold large amounts of equities. Habit investors are not good volatility strategy investors—when volatility spikes, prices tend to crash, and these are times when their wealth creeps closer to their habits. The volatility strategy loss comes at a particularly bad time.

4.5. CATCHING UP WITH THE JONESES

Catching up with the Joneses utility defines bad times relative to other investors. It is not your performance in absolute terms that matters; what matters is your performance relative to your peers. We say that utilities exhibit *externalities*

[23] Some important papers in this large literature include Sundaresan (1989), Constantinides (1990), and Campbell and Cochrane (1999).